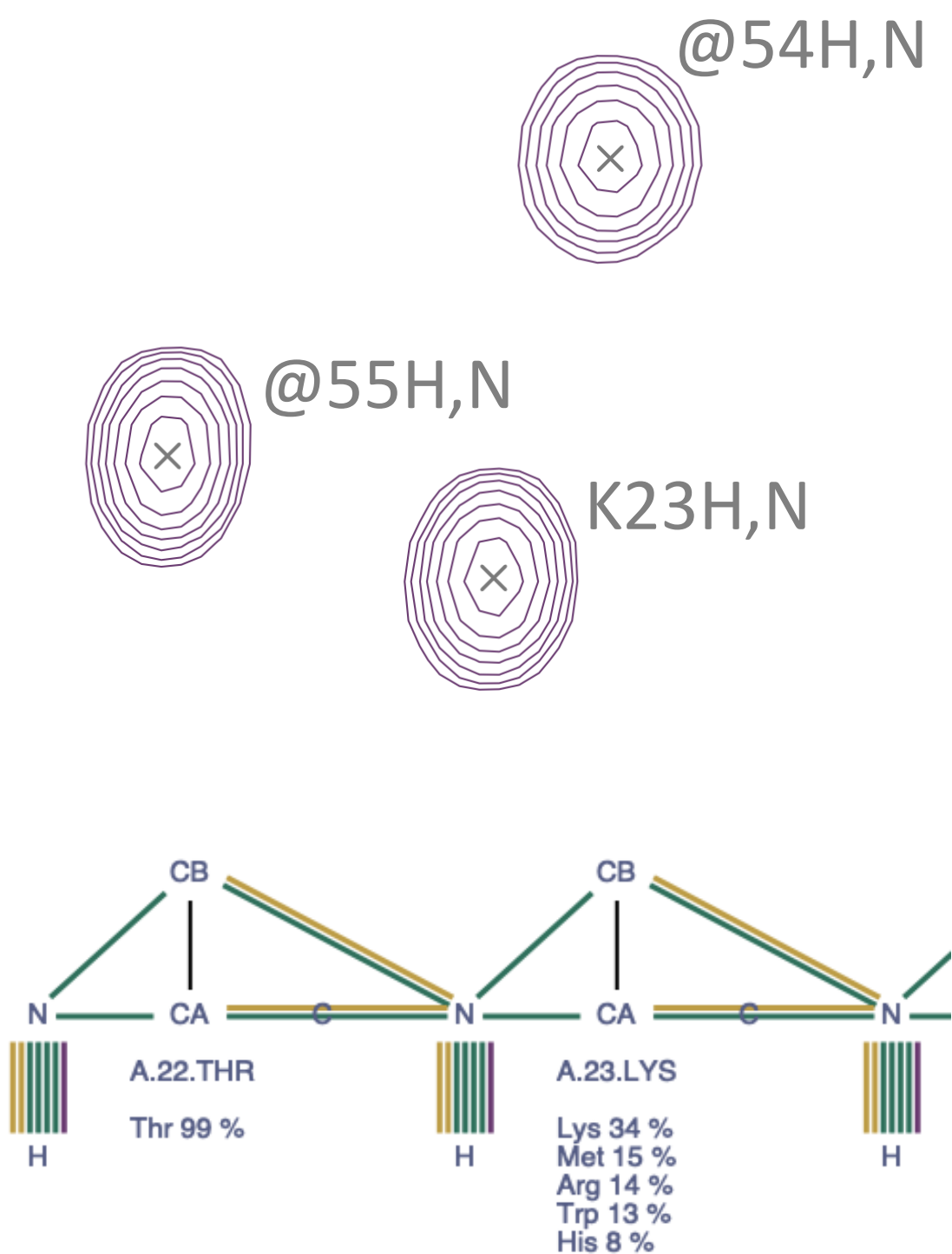


Backbone Assignment Tutorial



Introduction

These tutorials are designed to guide you through a sequential triple resonance backbone assignment using Ccpnmr AnalysisAssign Version 3.3, they are not intended to teach any theoretical aspects of NMR.

In these tutorials you will use spectra recorded on Sec5, in particular HSQC, CBCAcoNH, HNCACB, HNCA and HNcoCA spectra.

You will need four projects which are located in the **CcpnBBAssignTutorial** directory.

The first two projects, Sec5Part1 and Sec5Part2 are used for the actual sequential backbone assignment, the others are *How-To* examples of how to do common assignment operations using different AnalysisAssign tools.

Please note that the images shown are only representative and you may encounter minor differences to your setup.

Contents:

1. Project Setup
2. Picking Peaks
3. Setting up the Assignment
4. Peak Picking 3D Spectra
5. Sequential backbone assignment
6. Inspect the Assignment
7. Quick Assignment

Start CcpNmr Analysis V3

Apple users by double clicking the icon *CcpNmrAnalysis*



Linux users by using the terminal command:
bin/assign

Windows users by double-clicking on the
assign.bat file

Reminder: basic operations in CcpNmr Analysis

Sidebar

All data contained in a project, such as spectra and peak lists are located in the sidebar. **Double-clicking** on an item will open its properties popup. **Dragging** an item into the **Drop Area** will open it as a module (Spectrum Display, Table etc.)

Spectrum Display

A Spectrum Display can contain multiple overlaid spectra which share the same axes. To show/hide a single spectrum, click on its spectrum toolbar button. If you close a display, you can open a spectrum by **dragging and dropping** it into the drop area from the sidebar or by **right-clicking** on a sidebar item and selecting **Open as module**. You can also add additional spectra to a spectrum display module or drag several spectra into the drop area together to open them simultaneously.

Mouse

- Pan -> **Left-drag** in display
- Zoom in/out -> **Scroll wheel** in display
- Context menu -> **Right-click**
- Select a peak -> **Left-click** on a peak symbol "X"
- Move a peak -> select first, then **right-click and drag**

Two-Letter Shortcuts

Press the first letter on your keyboard e.g., **M**, followed by the second letter, e.g., **K** (case insensitive). Press **Esc** to cancel the first letter.

For more commands and operations

Main Menu → Help → Tutorials → Beginners Tutorial

OR

Main Menu → Help → Show Shortcuts

Main Steps

• Open project

- Drag in **Sec5Part1.ccpn**

• Pick HSQC Peaks

• Set up Assignment:

- Assign > Set up NmrResidues (**SN**)
- Sidebar > Chain > <New Chain> using protein name and sequence
- Spectrum > Set Experiment Types (**ET**)

• Pick and Assign 3D Spectra:

- Assign > Pick and Assign (**PA**) using Settings:
 - Tick **Automatic C/CA/CB NmrAtom assignment**
 - For **Pick Peaks** select **Display** and **3D_HCN** from drop-down
- **Double-click** on an NmrResidue in table to navigate
- Click **Restricted Pick and Assign** to pick peaks and assign H, N and C atom types for one or more rows at a time
- Correct carbon atom assignments, if necessary, using Assignment > NmrAtom Assigner (**AN**)

• Open project Sec5Part2.ccpn

• Sequential Assignment / Linking NmrResidues:

- Assign > Backbone Assignment (**BB**) using Settings:
Set **Match** and **Search** modules to different 3D Spectrum Displays
- Double-click on NmrResidue **@93-1** in BB Assignment table
- Drag NmrAtom of matching strip onto orange arrow in Search Spectrum Display:

GS:3D_HCN_1.1	C	GS:3D_HCN_1	GS:3D_HCN_1
NR:@-.@93	10	NR:@-.@53	NR:@-.@25
N:109.031[1]		[i-1] [97%]	[i-1] [75%]
		N:117.182[1]	N:123.823[1]

- Continue linking in i-1 direction
- For i+1 direction double click **@93** and drag onto right hand arrow

• Make Sequence Specific Assignment:

- In Sequence Graph: drag an NmrResidue onto matching Residue in Sequence (in lower part of module) to assign the whole stretch

• Inspecting / correcting Assignments:

- Find/correct mistakes in project **Sec5Part3.ccpn** using Assignment Inspector (**AI**), Peak Assigner (**AP**) and Sequence Graph (**SG**)

• Quick Assignments:

- Drag NmrAtoms or NmrResidues from sidebar onto selected peaks

1

2

3

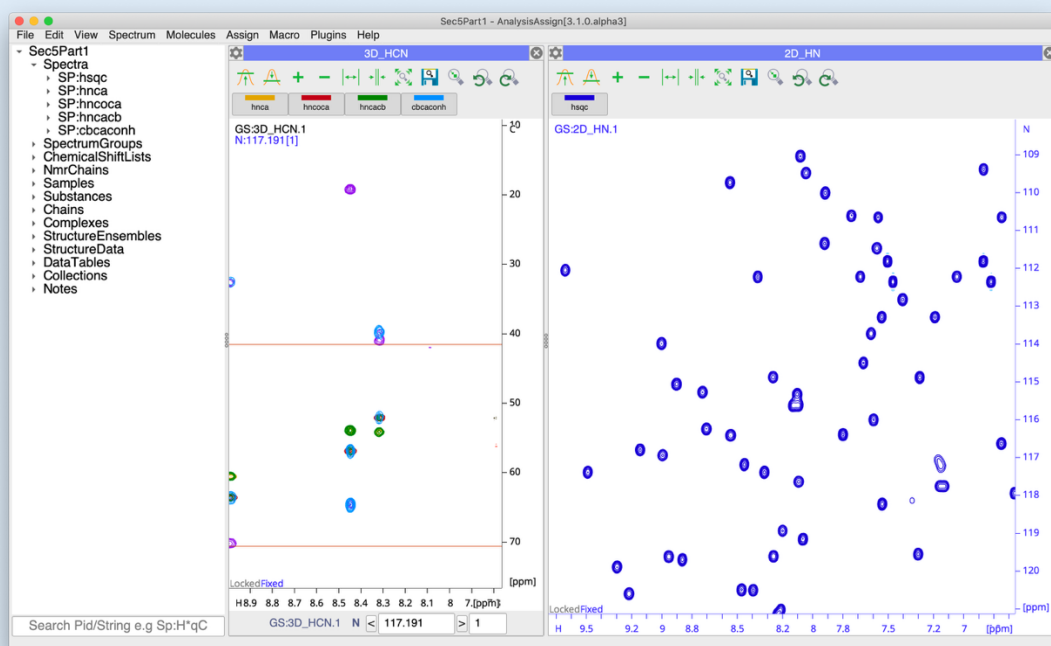
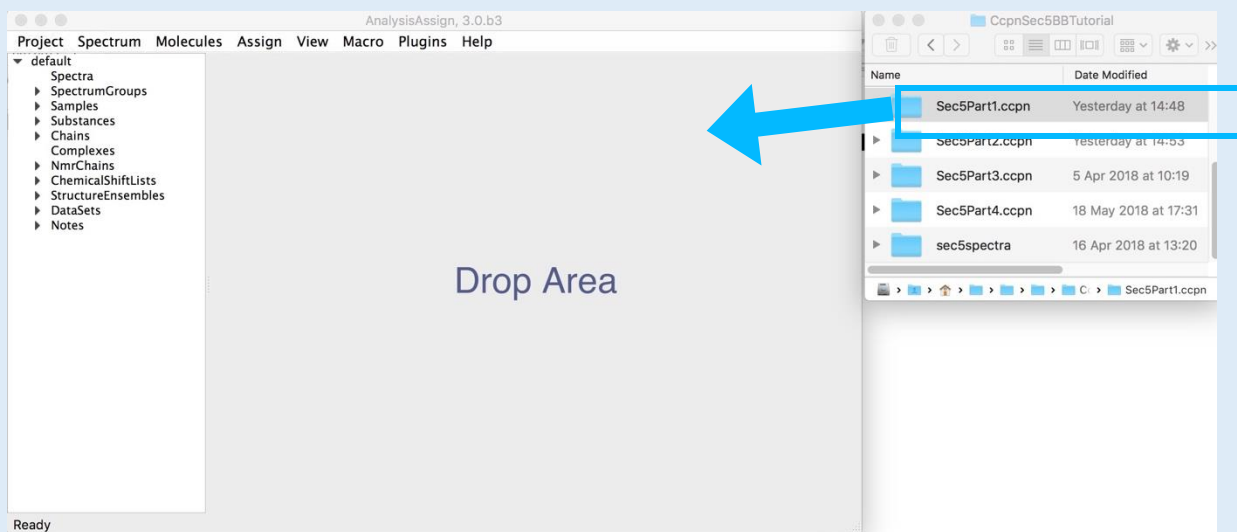
4

5

6

7

Open the project **CcpnBBAssignTutorial/Sec5Part1.ccpn**



1A Drag & drop Sec5Part1.ccpn into the sidebar or drop area.

CcpNmr projects are saved as directories with the **.ccpn**.

- Find the project directory **Sec5Part1.ccpn** in the **CcpnBBAssignTutorial** directory and drag & drop it into the program.

The Sec5Part1 project will be loaded in your program window.

You will see five spectra, displayed in two SpectrumDisplays as:

- hsqc (dark blue, 2D_HN)
- hnca (yellow, 3D_HCN)
- hncoca (dark red, 3D_HCN)
- hncacb (green/purple, 3D_HCN)
- cbcaconh (light blue, 3D_HCN)

The screenshot shows the 'Customise' menu on the left with 'Contours...' selected. A blue arrow points from the 'Contours...' option to the text 'or shortcut CO'. Below the menu, the 'SpectrumDisplay Properties' dialog box is open, showing the 'hncacb' tab. The 'Show Negative Contours' checkbox is highlighted with a blue box. A blue arrow points from the 'Contours...' menu option to the 'Show Negative Contours' checkbox. A blue box is also present in the bottom right corner of the dialog box.

Customise:

- Contours... C, O
- Calibrate Spectra
- Horizontal Trace T, H
- Vertical Trace T, V
- Enter Phasing-Console P, C
- Peak-Picking Mouse Mode M, M

Mark Position

Mark Axes:

Clear Marks

Navigate to:

Copy Axes:

Flip Axes:

Estimate Noise

Make Strip Plot

Auto-arrange

Reset All Labels

Selected Peaks

Export to File

SpectrumDisplay Properties

hncacb

Link Contours ☒

Show Positive Contours ☒

Positive Base Level 8.70175e6

Positive Multiplier 1.41

Number of positive contours 10

Positive Contour Colour goldenrod

Show Negative Contours ☐

Negative Base Level -8.70175e6

Negative Multiplier 1.41

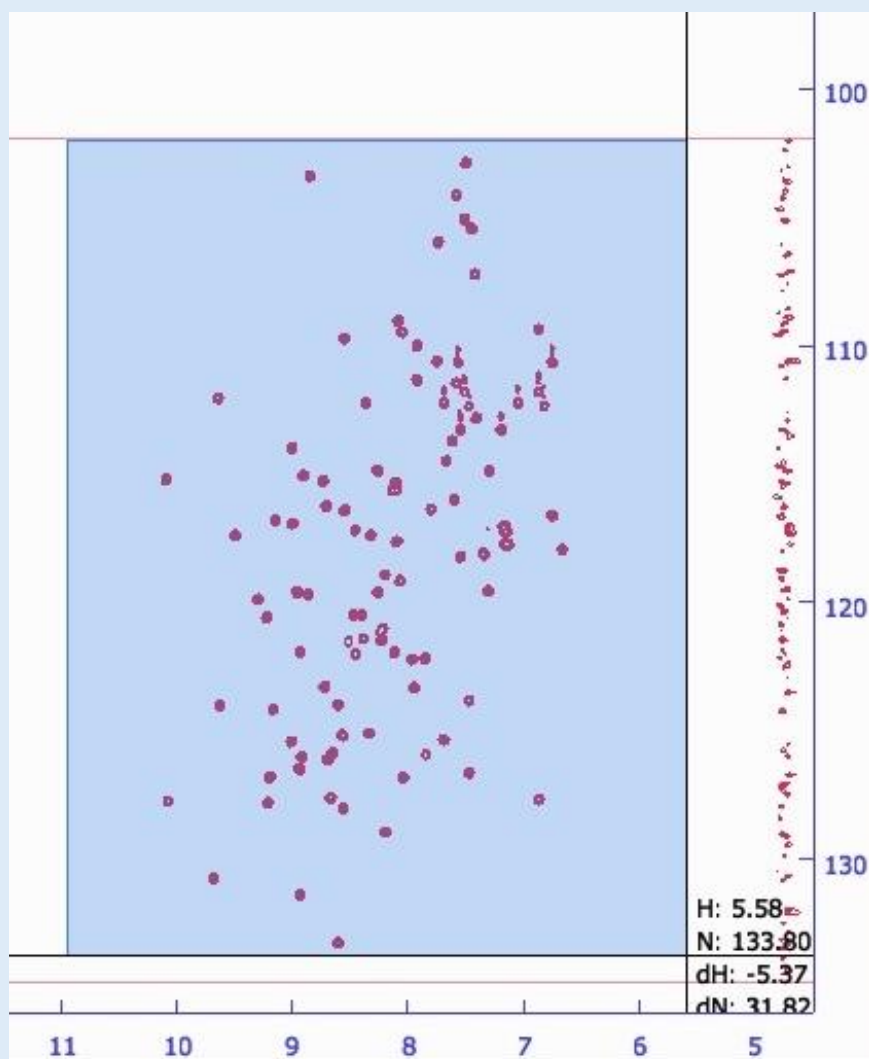
Copy Selected Attribute

If the contours are not displayed correctly, verify that the path is set correctly in the **Spectrum Properties** popup under the **General** tab options (you can open this by **double-clicking** on the spectrum in the sidebar). Spectra are located inside the **CcpnBBAssignTutorial/sec5spectra** directory.

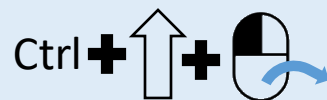
1B Set Spectrum Properties: Contours.

- **right-click** on each spectrum display and click **Contours...**
- uncheck **Show Negative Contours** except for the hncacb

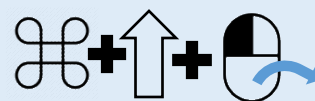
You can also perform these actions for each individual Spectrum in the **Spectrum Properties** popup; open them by **double-clicking** on the spectra in the sidebar.



Linux / Windows:



Mac:



2A Picking peaks in the HSQC

You can pick peaks either manually or automatically.

Manual Picking: Ctrl (or Cmd for Mac) + Shift + Left-drag

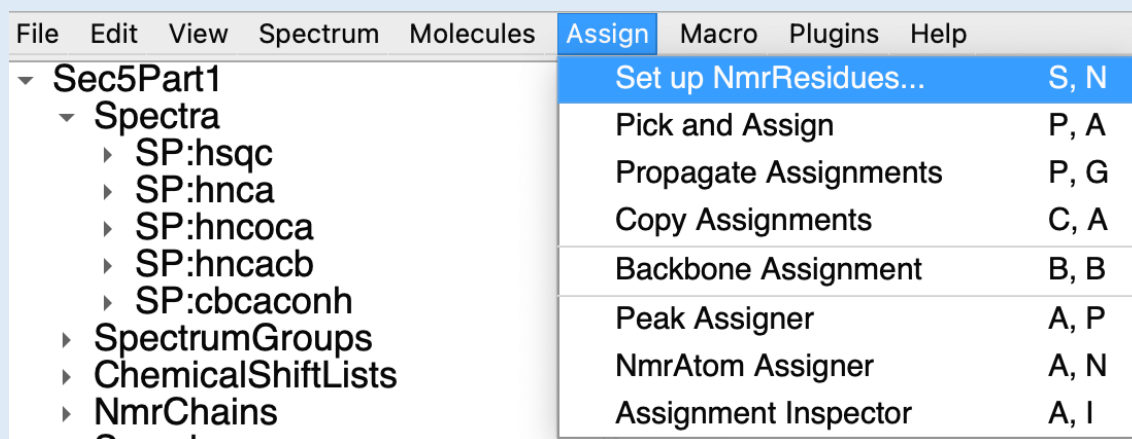
- Click on the spectrum display, hold down **Ctrl** (or **Cmd** for Mac) + **Shift** and **left-drag** the mouse to create a blue picking box in the regions:
 H → 11-6
 N → 102-134.50
- Then release the button and keys.

The crosses in the spectrum mark the peak positions picked.

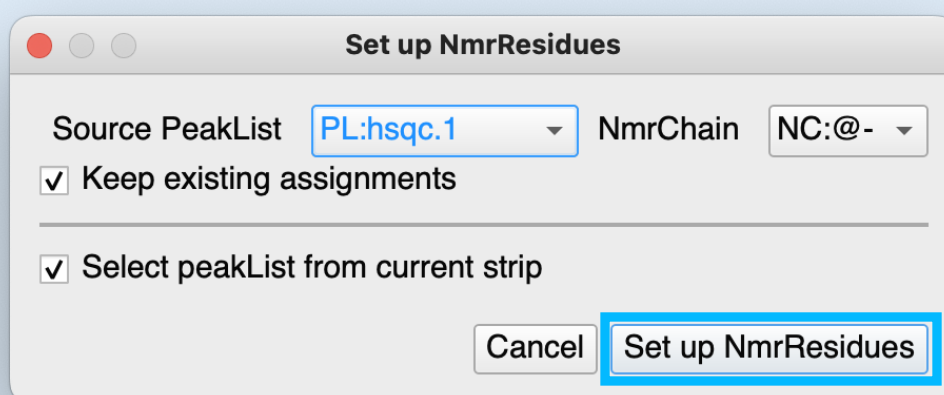
The hyphens separated by a comma indicate that the dimensions of these peaks are unassigned.



For more information about peak picking including other ways to pick your peaks, go to <https://www.ccpn.ac.uk/manual/v3/PeakPicking.html>



or Shortcut SN



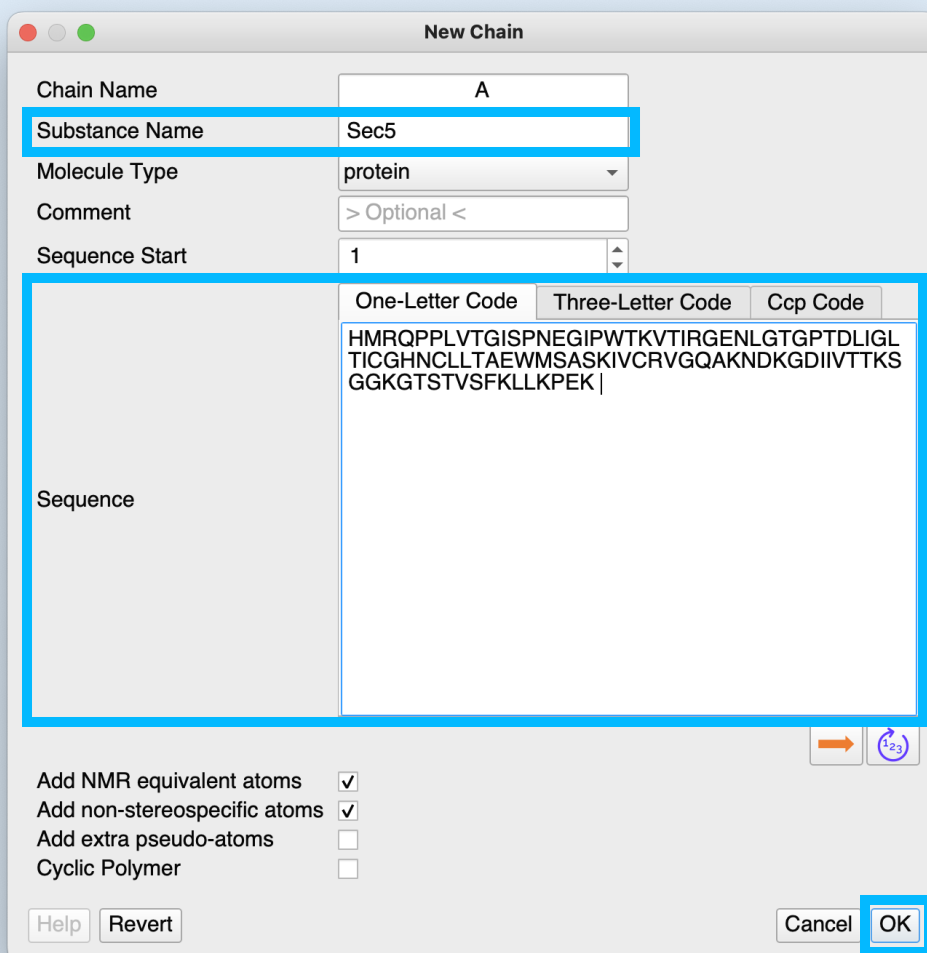
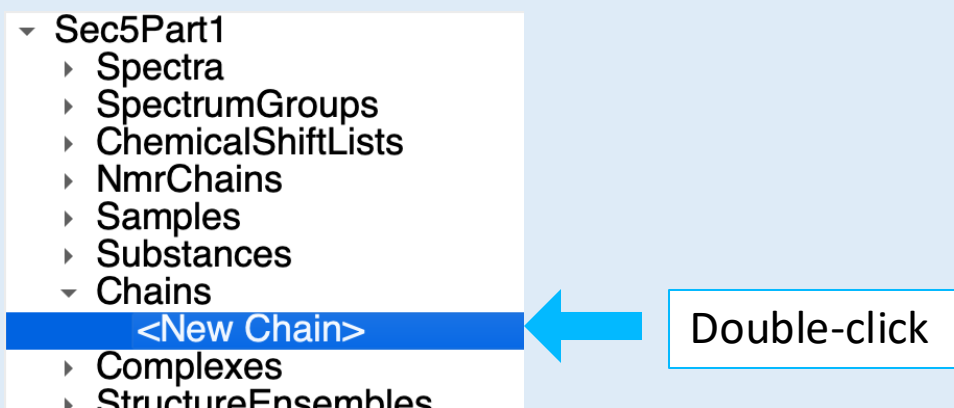
To start the backbone assignment process, we need to define 'anonymous' NmrAtoms, i.e. effectively random labels, for the two dimensions of the ^{15}N - ^1H HSQC peaks, which can subsequently be linked to the 3D spectra and used as placeholders until the actual assignments are obtained. See page 9, at the end of Point 3, for a full explanation of the AnalysisAssign nomenclature.

3A Setup NmrResidues

- Go to **Main Menu** → **Assign** → **Setup NmrResidues** (or use shortcut **SN**)
- Source PeakList: **PL:hsqc.1**
- NmrChain: **NC:@-**
- Click **Set up NmrResidues**

Each peak will now have a label such as **@1H,N**. Use the shortcut **PL** to toggle between different ways to display the peak labels or go to the **SpectrumDisplay Settings** (click on the gear icon in the top left hand corner) to set the way you wish to view your PeakLabels. There are also further settings if you **double-click** on the PeakList in the sidebar.





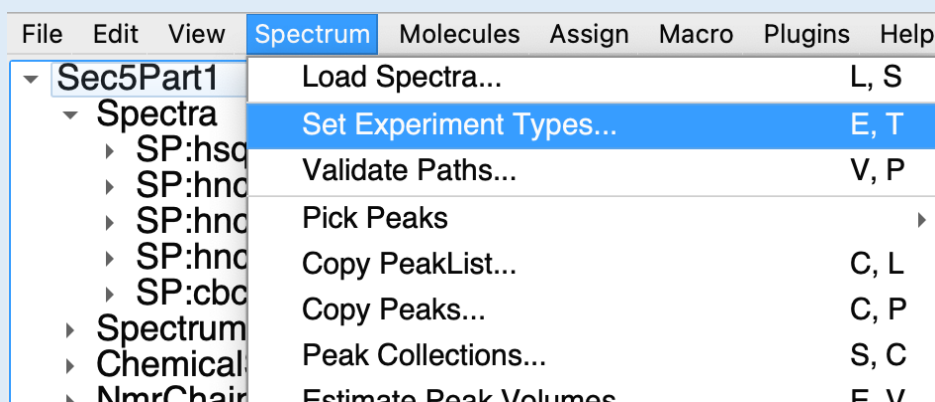
3B Create Chain

- Go to **Sidebar** → **Chains** → **<NewChain>**

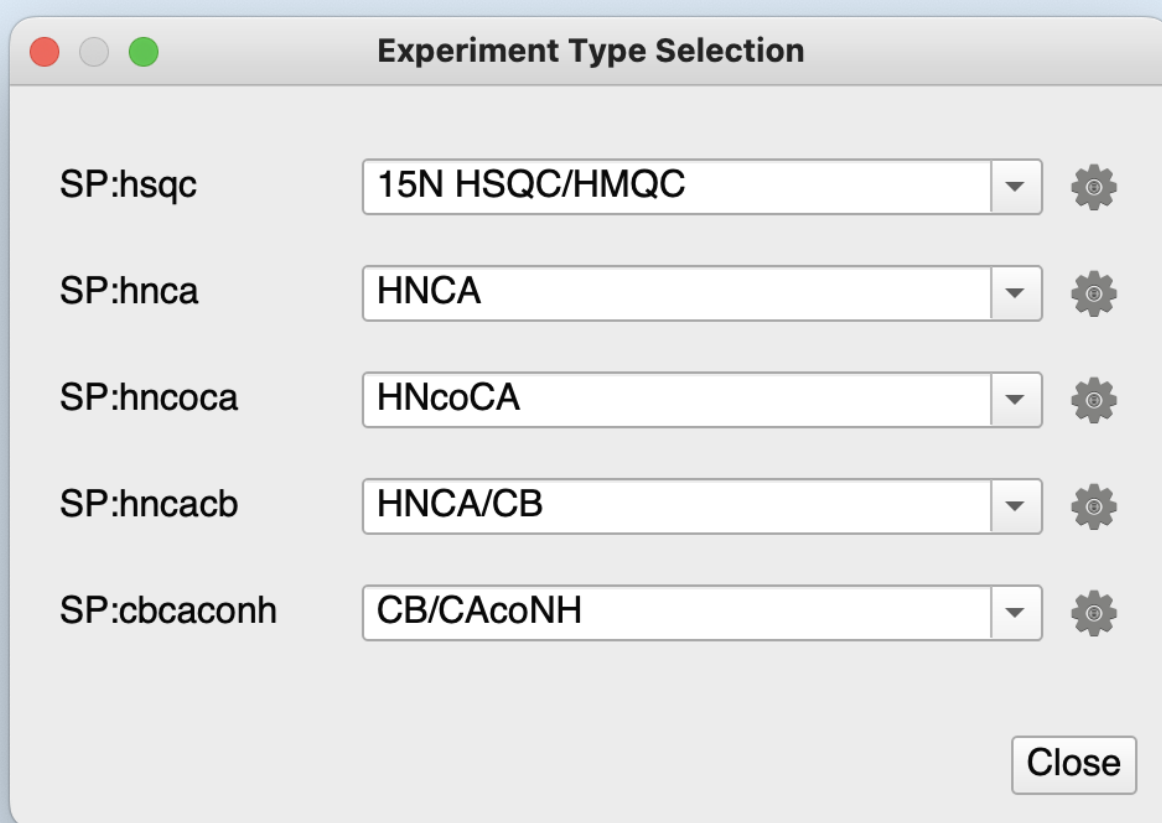
Use the default settings with

- Substance Name:** Sec5
- Sequence:** (copy and paste)
 HMRQPPLVTGISPNEGIPWTKVTIRGENLGTGPTDLIGLTICGHNCLLTAEWMSASKIVCRVGQA
 KNDKGDIIVTTKSGGKGTSTVSFKLLKPEK
- Click **OK**

You can also use **Main Menu** → **Molecule** → **Generate Chain** or drop a FASTA formatted file into the project.



or shortcut **ET**



3C Set Experiment types

It is possible to tell the programme what kind of experiment each spectrum is. Although this is not strictly necessary in order to assign you spectra, it will allow you to make use of several helpful features later on.

- Go to **Main Menu** → **Spectrum** → **Set Experiment Types...** or shortcut **ET**
- Set the Experiment Types from the drop-down menus as shown above and then click **Close**.

Assignment nomenclatures (Explanation only)

Assignment in AnalysisAssign is simply a matter of setting strings that define the NmrAtoms.

We call this the 'id' (id: identifier) of the NmrAtom. If an id matches the strings defining a molecular Atom¹, this effectively constitutes an assignment to the Atom.

If not, the NmrAtom id is a placeholder, reflecting its progress towards assignment².

At this point, it is appropriate also to consider the relationships between Peak, ChemicalShift and NmrAtom. Each dimension of a Peak is assigned to one or more NmrAtoms. The ChemicalShift (which resides in a ChemicalShiftList) of an NmrAtom, is defined by all the peaks that have been assigned to this NmrAtom. Hence, changing an assignment for a Peak (e.g. reassigning a peak from "nmratom_1" to another "nmratom_2") has an effect on the ChemicalShift of "nmratom_1", as it is now no longer defined by the Peak. Likewise, it also affects the ChemicalShift of the "nmratom_2", as it now comes to be (also) defined by the Peak. We will see in next sections how to inspect and change the assignment(s) of a Peak. If you change the id of an NmrAtom (or its parent NmrResidue or NmrChain), the assignment of all ChemicalShifts and Peaks are updated.

We use NmrChains and NmrResidues to keep track of the NmrAtoms during the assignment process. By default, new NmrResidues are put in NmrChain '@-', and new, temporary NmrChains are given names like '@2'. Initially, NmrChains contain no information about the sequential connections of the NmrResidues, i.e. their ordering. In this case, the NmrChain functions like a simple list with all its NmrResidues. To store sequential stretches, i.e. lists in which the NmrResidues are ordered, the program uses 'connected' NmrChains, whose names start with '#' instead of '@'. Consequently, names with '@' (and NmrChain names starting with '#') are reserved.

NmrResidues are created with names like '@173' and with no residueType. When you want to create the previous, 'i-1' residue to a given residue (for backbone assignment) you give it a '-1' suffix, in this case '@173-1'. When you assign the NmrResidue to a real residue, renaming it e.g. 'A.45.GLY', the i-1 residue name updates to 'A.45-1'.

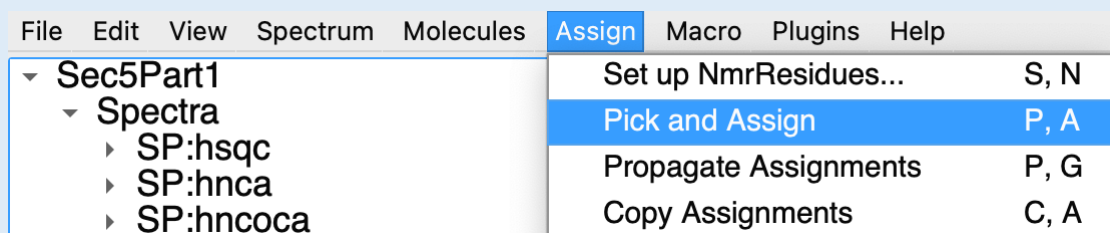
NmrAtom names always start with the nucleus, and default names would be e.g. 'H@31' or 'C@88'. Some names have a special meaning:

- '%' means 'any number', so 'HB%' would be a beta methylene or methyl group. 'H%' would be the backbone NH3 group.
- '*' means 'any string', so 'H*' would be 'any proton in the residue'
- Names starting with 'M' and 'Q' are (proton) pseudoatom names
- Number suffixes follow NEF (IUPAC) convention, so serine HB2 or HB3 denote stereospecific assignments.
- Suffixes 'x' and 'y' are used for non-stereospecific pairs - the normal assignment to serine beta would use HBx and HBy. For e.g. isopropyl groups the x and y assignments match up between ¹H and ¹³C so that Leu HDx% are the methyl protons bound to Leu CDx (NEF convention).

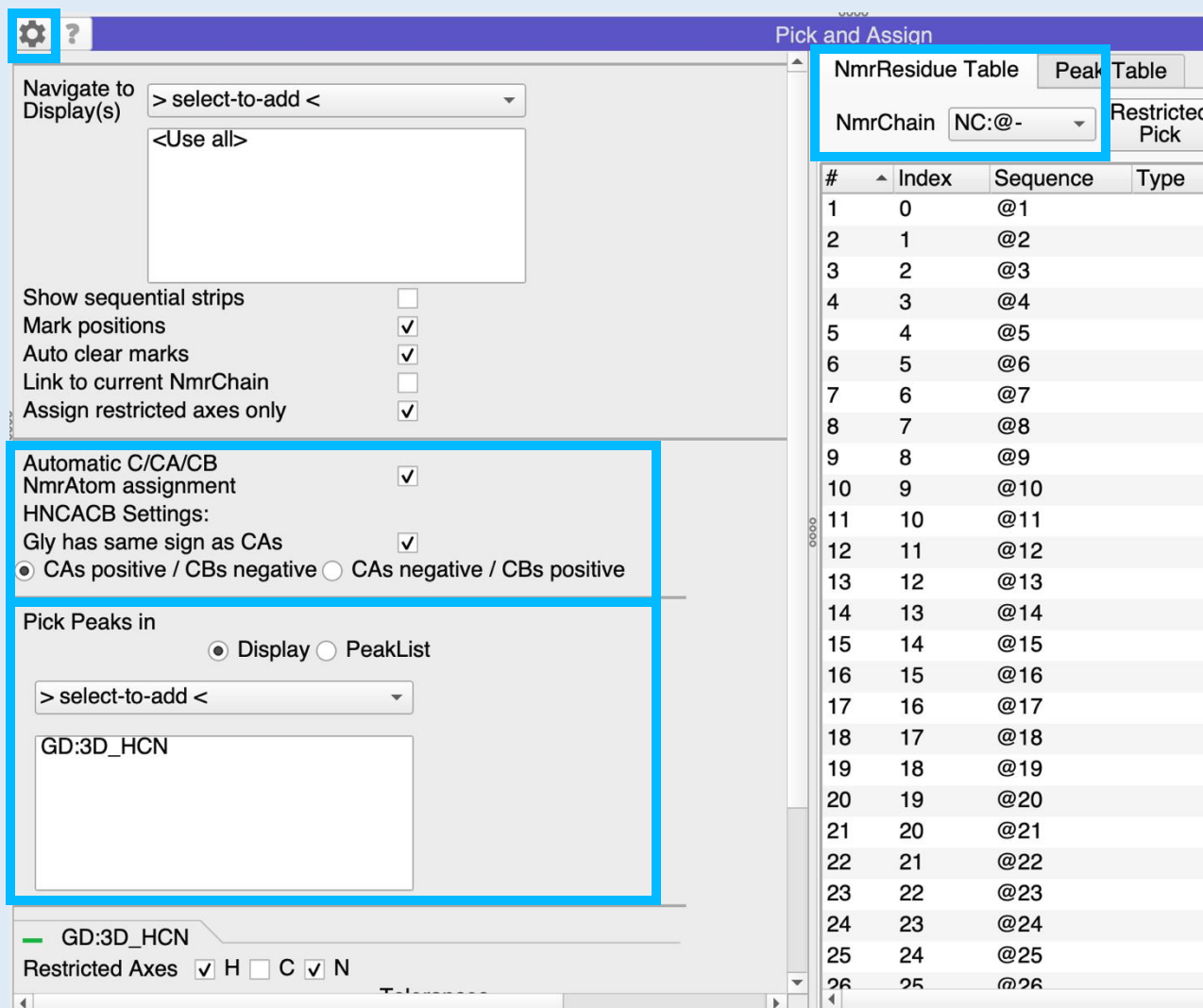
¹ Atoms reside in Residues, which reside in Chains; multiple chains can form a Complex.

² The id together with the type identifier forms the so-called pid, the project-identifier. As an example for an un-assigned amide in the 123rd NmrResidue in the second NmrChain: NA:@2.@123..H. For an assigned NmrAtom, all the fields will have been filled, yielding something like NA:A.GLU.14.H.

For more information see our video tutorial on NmrResidues at <https://youtu.be/DS9IZzNsBbQ>



or shortcut **PA**

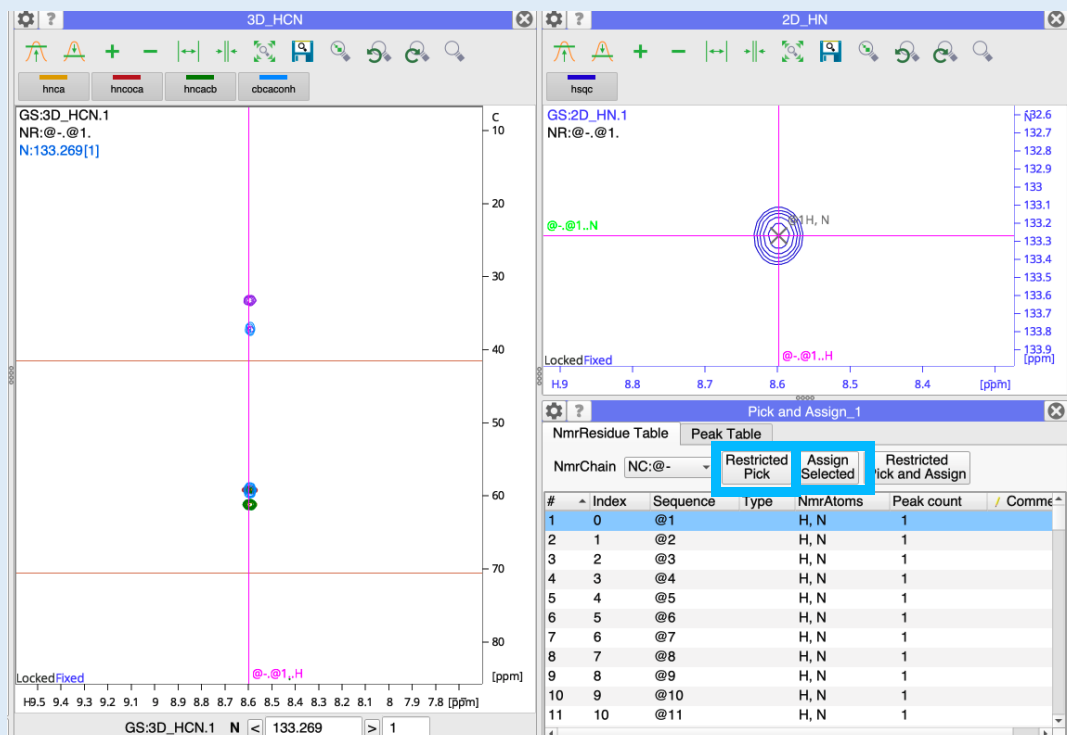


4A Pick and Assign 3D Peaks Setup

We will pick the 3D spectra based on the HSQC spectrum using the **Pick and Assign** module:

- Go to **Main Menu** → **Assign** → **Pick and Assign**, or shortcut **PA**.
- Open the Settings (gearbox icon ) and select:
 - Automatic C/CA/CB NmrAtom assignment** (using the options shown above)
 - Pick Peaks in Display **3D_HCN**
 (if required, remove **<Use All>** by **right-clicking** and selecting **Remove**)
 - Leave the remaining options as default and close the settings
- Ensure the selected NmrChain is **NC:@- .**

Arrange your modules with the **Pick and Assign** module below or above the **2D_HN** (HSQC containing) module:



4B Pick and Assign 3D Peaks

In the Pick and Assign module:

- **Double-click** on a row in the table, e.g. the row for @1.

This will cause the spectrum display with the HSQC spectrum to focus on the position of the peak labelled @-@1..H,N and to mark this position with two labelled rulers corresponding to each dimension. Likewise, the **3D_HCN** module navigates to the corresponding z (i.e. N) position and marks the appropriate frequency along the proton axis with a labelled ruler.

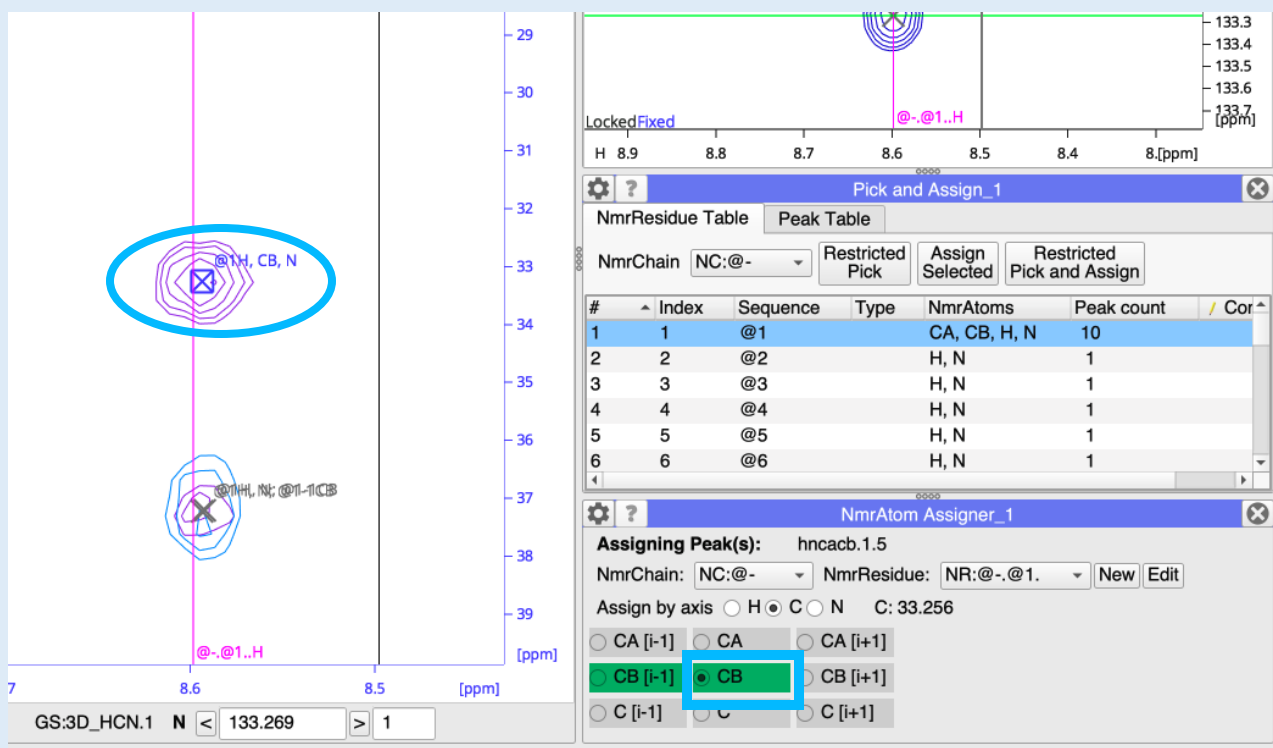
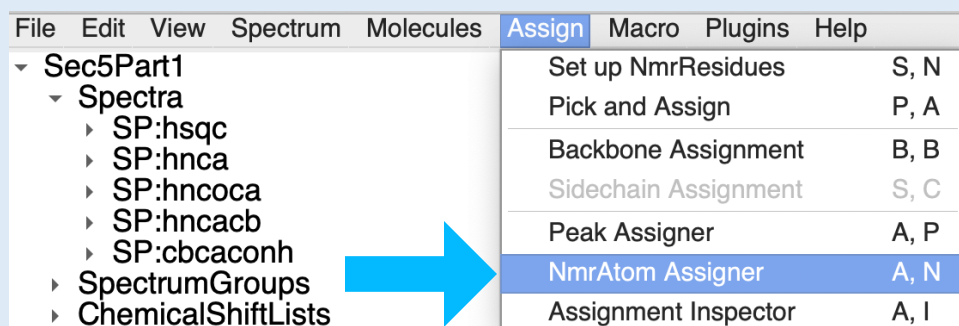
- Click the **Restricted Pick** button in the **Pick and Assign** module.

This will pick all the peaks along a narrow region in the 3D window and these peaks will be selected.

- Inspect the 3D peaks and delete any noise peaks.
Be aware that the peak picking occurs on any visible peaks, so it can be helpful to adjust the contours levels to something suitable before doing the **Restricted Pick**.
- Select the peaks to assign and click the **Assign Selected** button in the **Pick and Assign** module.

This transfers the H and N atom assignments from the 2D spectrum to the 3D spectra and assigns the Carbon NmrAtoms.

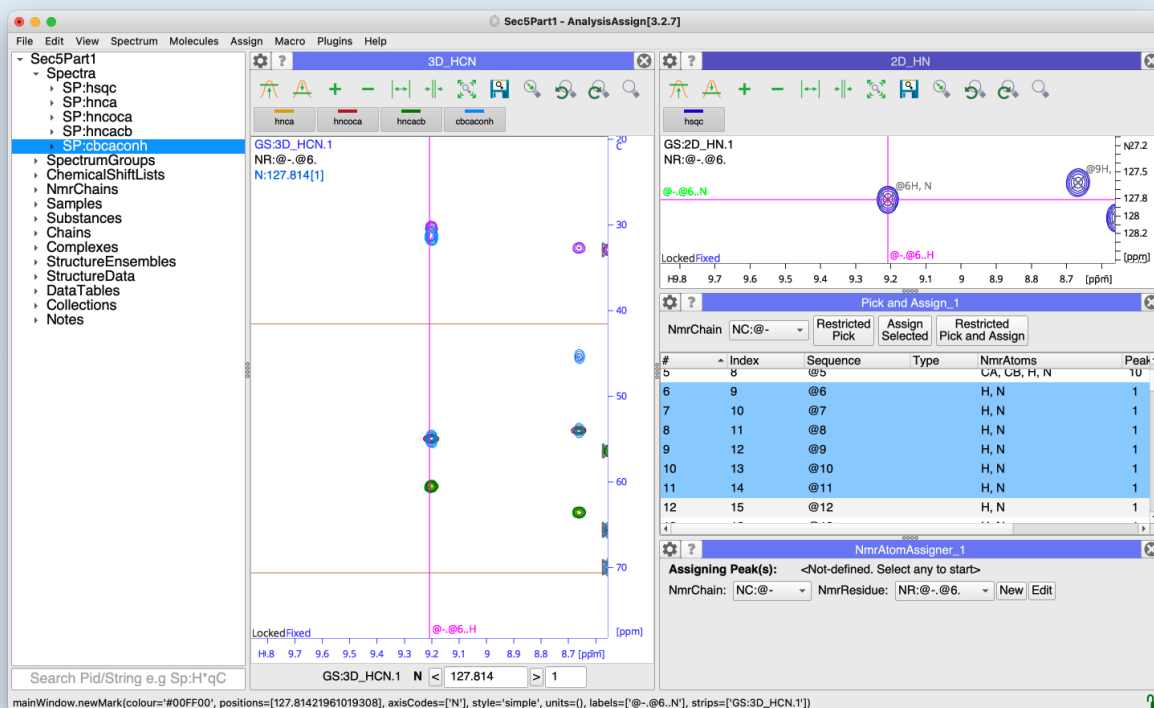
or
shortcut
AN



4C Correcting carbon atom type assignments

Occasionally you may find that the Carbon atom types have been assigned incorrectly (e.g. if peaks are overlapped, too many peaks have been picked or the peak intensities are unusual). In this case you can make corrections using the **NmrAtom Assigner** module:

- Go to **Main Menu** → **View** → **NmrAtom Assigner** or use the shortcut **AN**.
- Select one or more peaks at a single position in the **3D_HCN** window
The NmrAtom Assigner will predict the assignment for the carbon dimension, using green for likely and orange for less likely assignments.
- Toggle any NmrAtom button to assign/deassign it to the selected peak(s).



4D Continue to Pick and Assign the 3D spectra

To use the sequential Backbone Assignment tools in AnalysisAssign, the *i* and *i*-1 assignments for all NmrResidues need to be provided, so the procedure described above should be carried out for all NmrResidues listed in the Pick and Assign table, where possible.

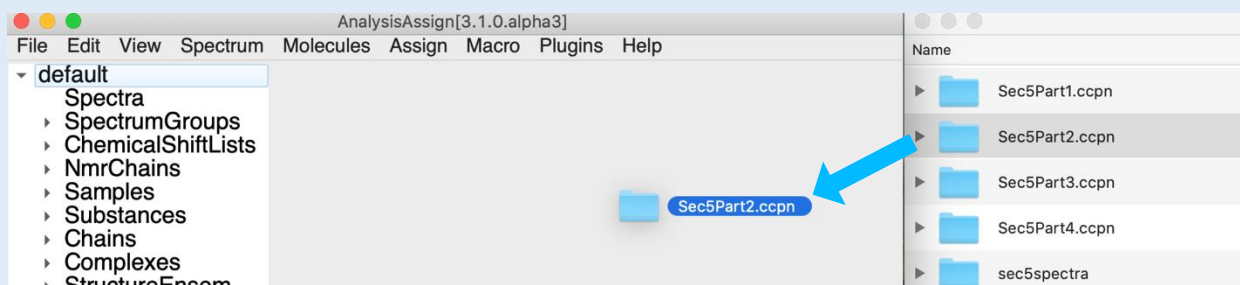
If you move on to assign NmrResidue **@2**, after double-clicking you will see no peaks in the **3D_HCN** spectral display; this NmrResidue originates from a Tryptophan side-chain NH and therefore does not display any peaks in the triple-resonance spectra. Move on to NmrResidue **@3** to continue, and so on for a few more residues to get the hang of it.

You can also Pick and Assign multiple rows (even the whole table) in one go:

- Use the **Shift** or **Ctrl/Cmd** key to select multiple rows.
- Click on **Restricted Pick** or **Restricted Pick and Assign** to pick and/or assign the peaks for all selected NmrResidues in one go.

It is a matter of personal preference whether you want to go through each strip and make sure it has been correctly picked and assigned, or whether you just pick up on any errors later as you make your assignments.

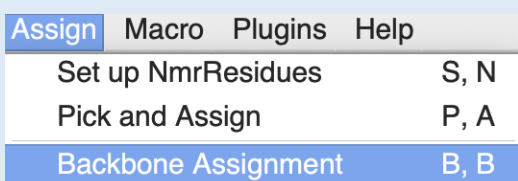
Once all three dimensions of all 3D peaks have been assigned to the appropriate NmrAtoms, the backbone assignment can be carried out, but obviously for this tutorial we provide you with a CcpNmr project in which this has already been completed.



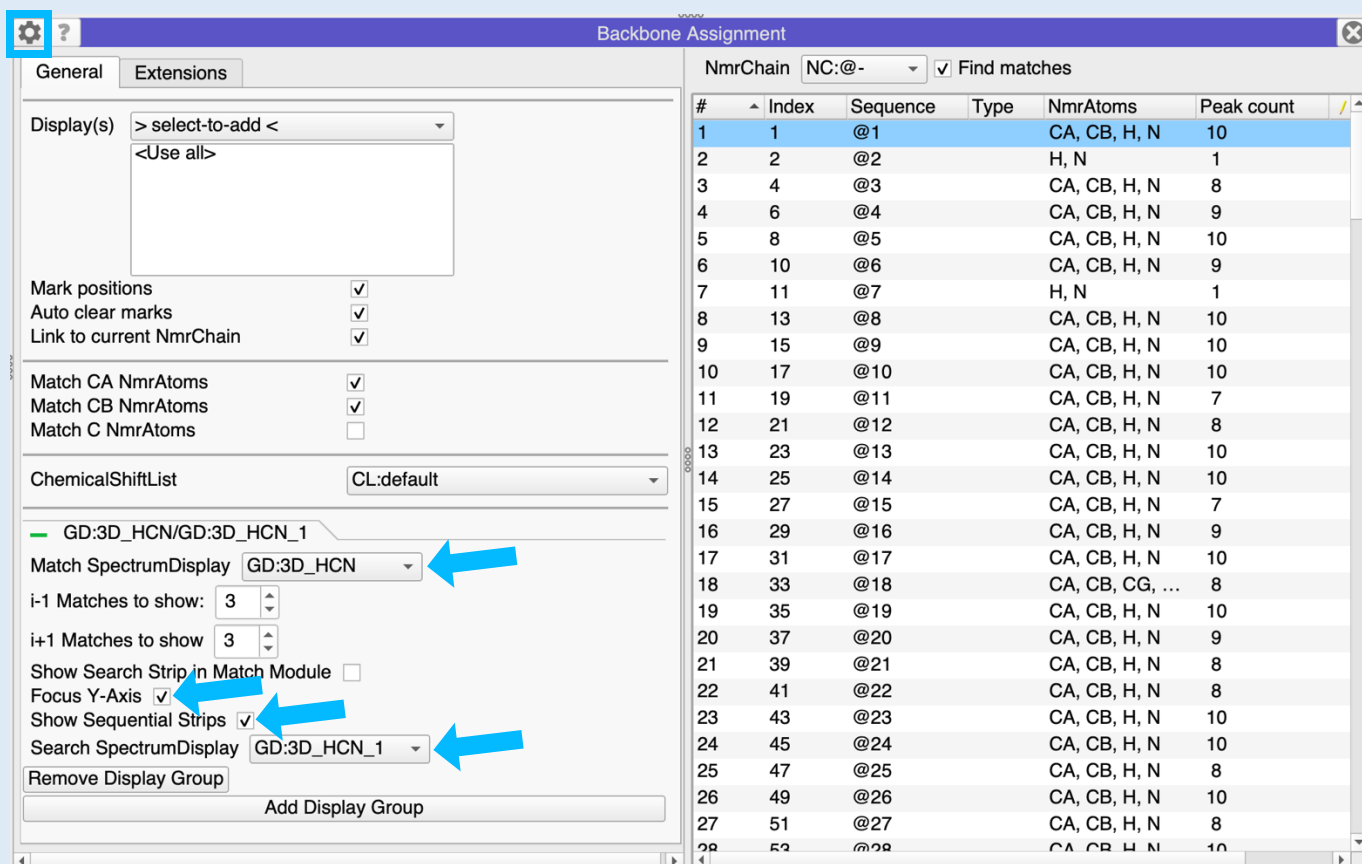
Sec5Part2.ccpn (in the **CcpnBBAssignTutorial** directory) is a project in which all the carbon atom type assignments for the NmrResidues have been completed and thus can be used directly for the sequential backbone assignment.

5A Open Sec5Part2.ccpn

- Open the **Sec5Part2.ccpn** project by dragging and dropping it into the **Drop Area** or on the sidebar. Quit the current **Sec5Part1** project with or without saving.



or shortcut **BB**

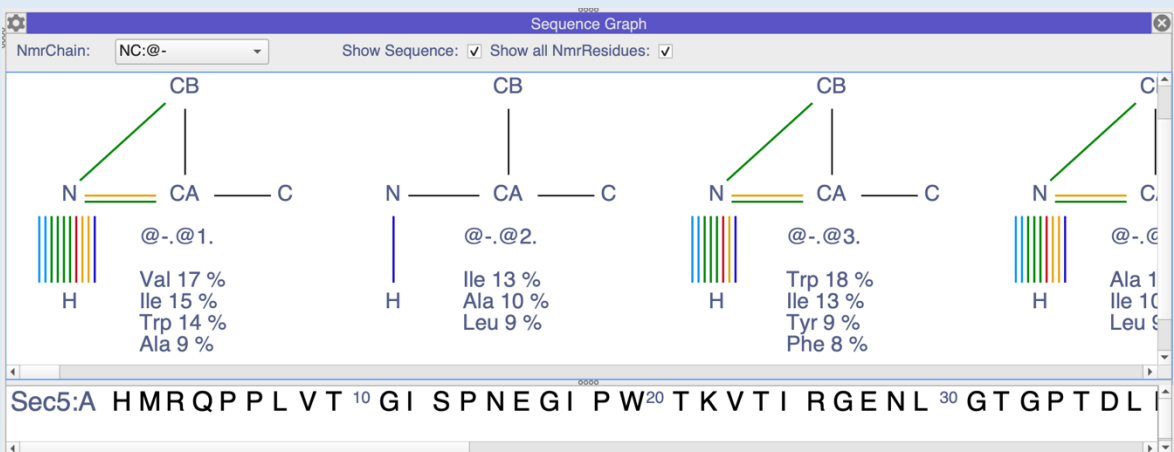


5B Open and set up the backbone assignment module

- Main Menu** → **Assign** → **Backbone Assignment** or shortcut **BB**
- Select the NmrChain: **NC:@-**.
- Open the settings (gearbox icon) and select:
 - Match module: **3D_HCN** (or similar 3D SpectrumDisplay)
 - Search module: **3D_HCN_1** (a different 3D SpectrumDisplay)
 - Tick **Focus Y-Axis** and **Show Sequential Strips**

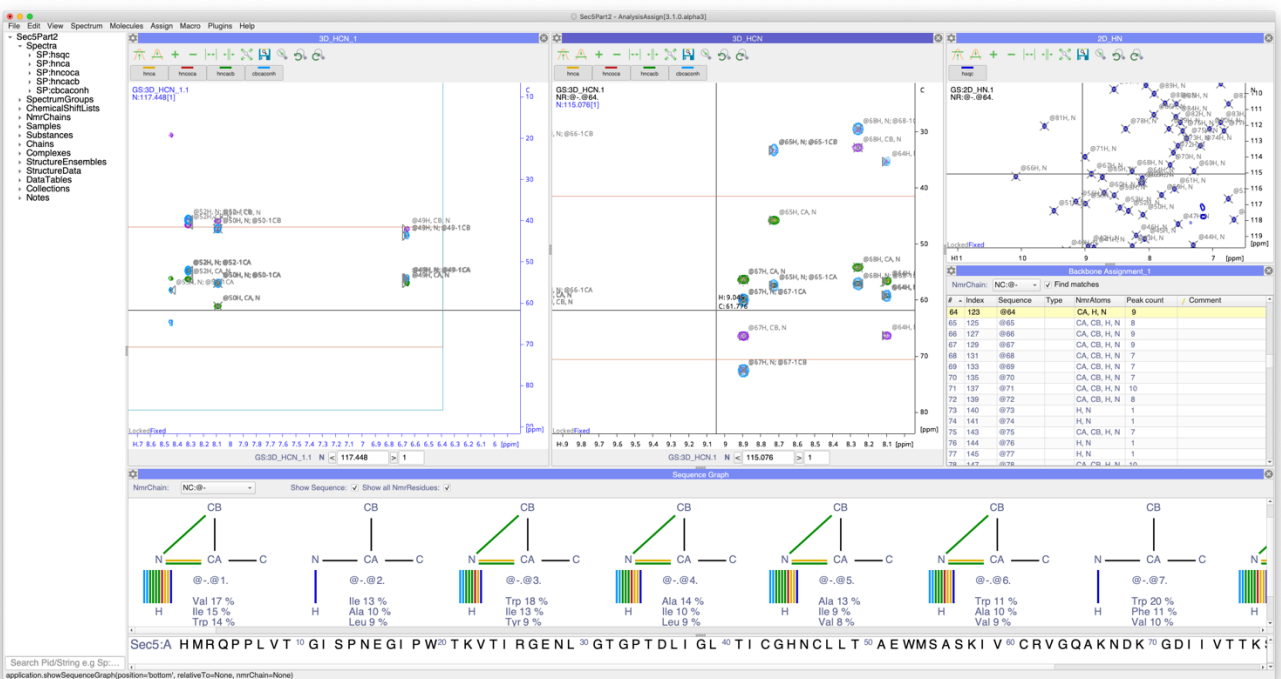
View	Spectrum	Molecules	Assign	Macro	Plugins
Chemical Shift Table				C, T	
NmrResidue Table				N, T	
Residue Table					
Peak Table				P, T	
Integral Table				I, T	
Multiplet Table				M, T	
Restraint Table				R, T	
Structure Table				S, T	
Data Table				D, T	
Sequence Graph				S, G	
Violation Table				V, T	

or Shortcut **SG**



5c Open and set the Sequence Graph module

- Main Menu → View → Sequence Graph or shortcut **SG**
- Select the NmrChain: NC:@-
- Rearrange the modules to a layout like the Figure below.



Note: If, using your own spectra, you do not see the coloured lines in the Sequence Graph, you will need to go to your **Spectrum Properties** and make sure your **Reference Experiment** and **Reference Experiment Dimensions** have been correctly set in the **Dimensions** tab.

Backbone Assignment_1

NmrChain NC:@- ☒ Find matches

#	Index	Sequence	Type	NmrAtoms
1	1	@1		CA, CB, H, N
2	2			H, N
3	4			CA, CB, H, N
4	6			CA, CB, H, N
5	8			CA, CB, H, N
6	10			CA, CB, H, N
7	11			CA, CB, H, N

or
shortcut **FT**

Right click in table

Backbone Assignment_1

NmrChain NC:@- ☒ Find matches

#	Index	Sequence	Type	NmrAtoms
180	169	@93-1		CA, CB

Equal

Filter in <Visible Table>

Double-click on row **@93-1**

5_D Assign in the i-1 direction ...

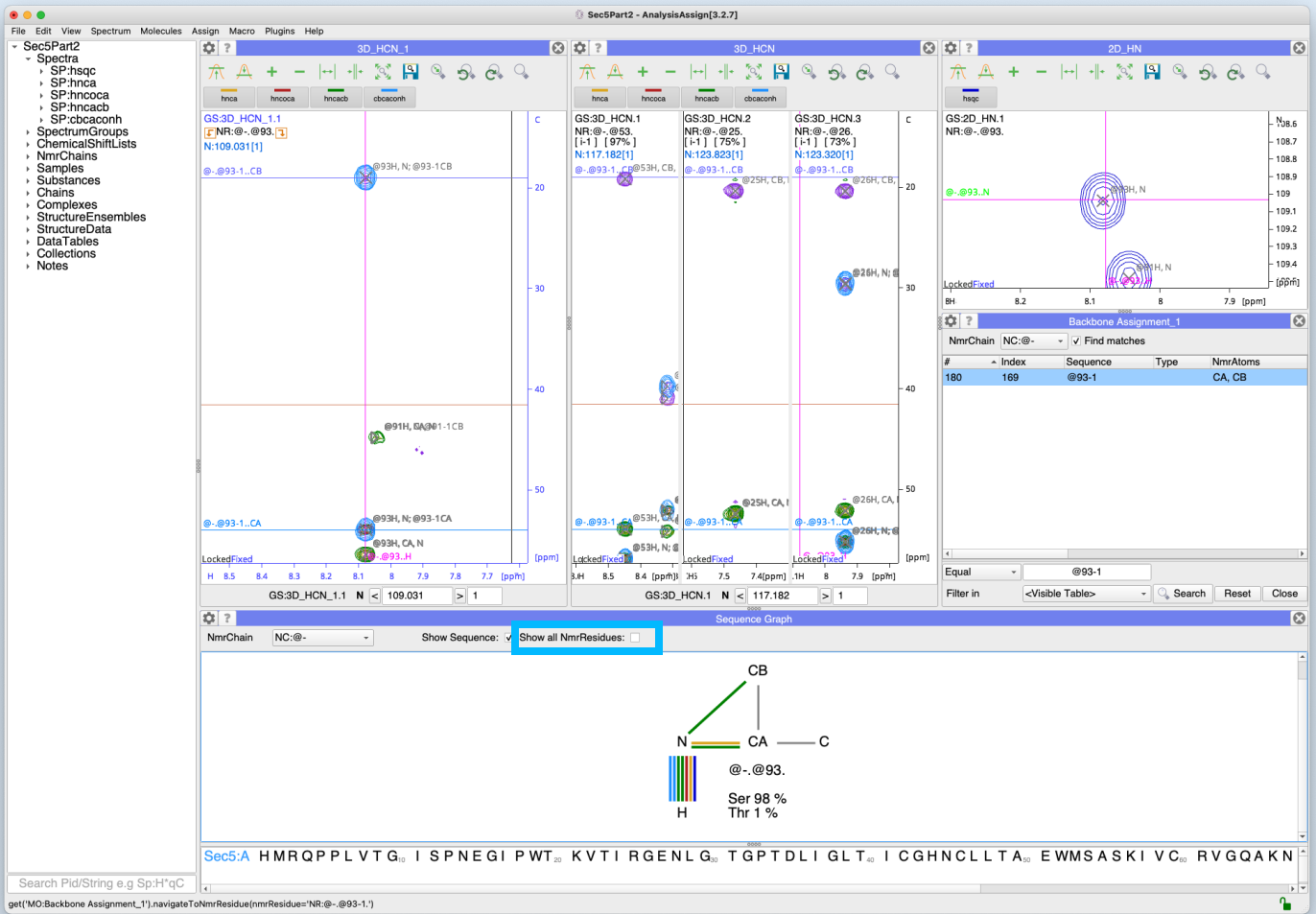
- Find the row for NmrResidue **@93-1**

You can do this by filtering for the string **@93-1** if you wish:

- Right-click** on the Backbone Assignment module table and click on **Filter...** or use shortcut **FT**
- search for **@93-1**
- Double-click** the **@93-1** row. You will see a series of changes in the GUI.

5 Sequential backbone assignment

Sec5Part2



5D ... Assign in the i-1 direction continued

The **3D_HCN_1** module will navigate to the appropriate plane containing the assignment for the NmrResidue **@93**; the relevant frequencies of **@93** H and N and **@93-1** C atoms are marked in all SpectrumDisplays.

- If you deselect **Show all NmrResidues** in the SequenceGraph:
Then **Sequence Graph** will draw a schematic residue, labelled with the NmrResidue name (**@93**) and predictions of the possible residue type(s) below it. The match module (**3D_HCN**) will display three strips in order (left to right) that the AnalysisAssign algorithm thinks best match the i-1 chemical shifts of **@93** (C-terminal of the residue **@93**).

Tip: Setting the 3D SpectrumDisplays as **Fixed** or **Locked** will keep your peaks in a nice shape.



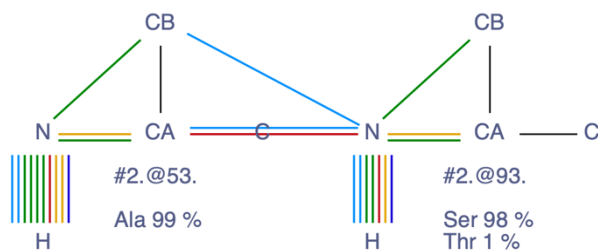
Drop on

Select and drag **NR:@-.@53.**

5E Connecting NmrResidues

Upon examination, the first strip in the Match module Spectrum Display (3D_HCN) shows NmrResidue **@53**; the H, CA and CB marks align with the peaks of this residue, we can therefore see that is the best match for **@93-1**.

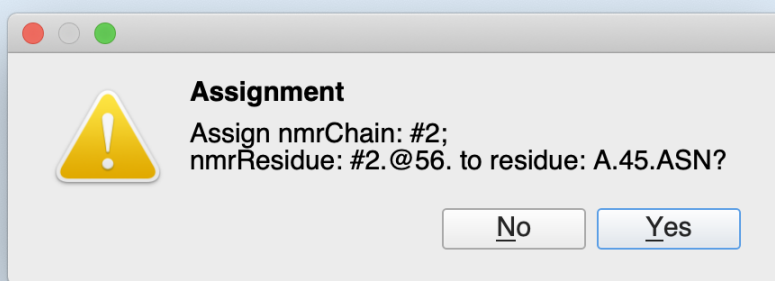
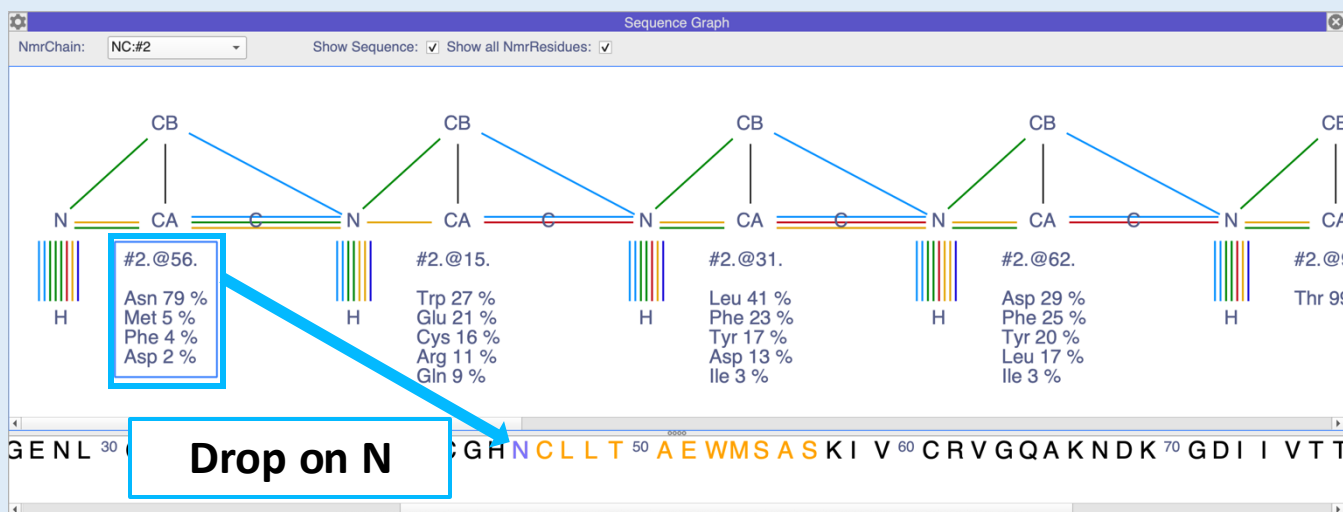
- Select the strip label **NR:@-.@53.** on the Match SpectrumDisplay
- Drag and drop the label on to the  in the Search SpectrumDisplay
- Check **Show all NmrResidues** in the **Sequence Graph** module (and make sure NmrChain **#2** is selected) to see the new connection



At this point **@93** and **@53** have been put into a so-called connected stretch (cf. section 3) and they are in a new NmrChain called **#2**, which means that their names have changed to **#2.@93.** and **#2.@53.** The NmrChain pulldown in the module should automatically be set to **#2**.

When strips are dragged and dropped through this procedure, AnalysisAssign will subsequently look for i-1 matches for **@53**. The algorithm thinks that **@70** is a good match for **@53-1** and on inspection it is a match, so selecting and dragging **@70** onto the  of the Search Spectrum Display will continue the assignment.

Note: The Settings panels of both the **Sequence Graph** and **Backbone Assignment** modules contain a setting **Link to current NmrChain**. If this is checked, then selecting a new NmrChain in one module will automatically update that used the other module. You can switch this on or off as suits you best.



5F Link NmrChain to Chain

If you continue assigning in the i-1 direction, you should end up with a stretch consisting of:

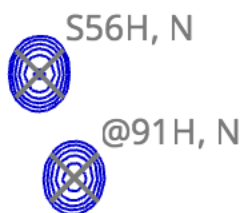
@40 @56 @15 @31 @62 @96 @25 @23 @6 @19 @70 @53 @93

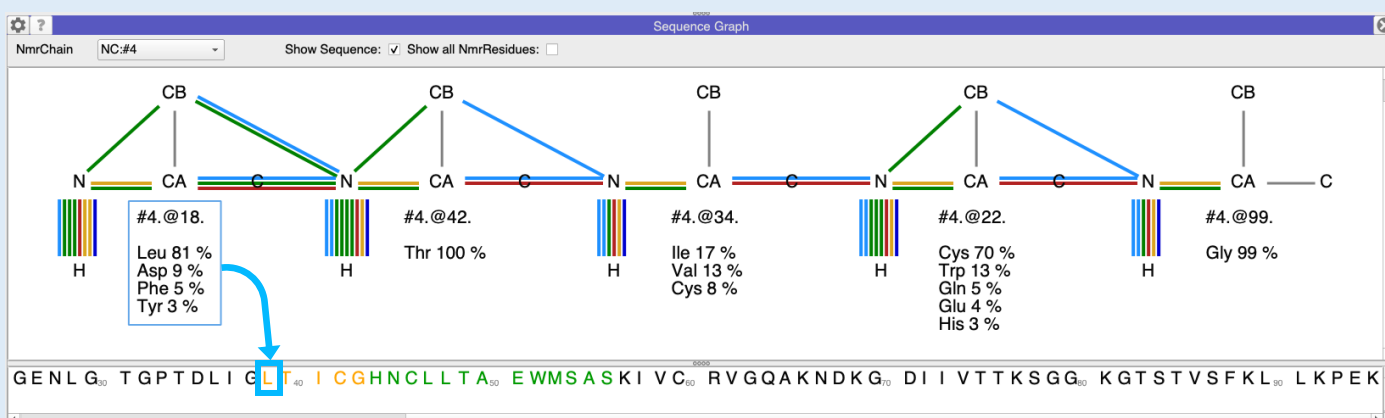
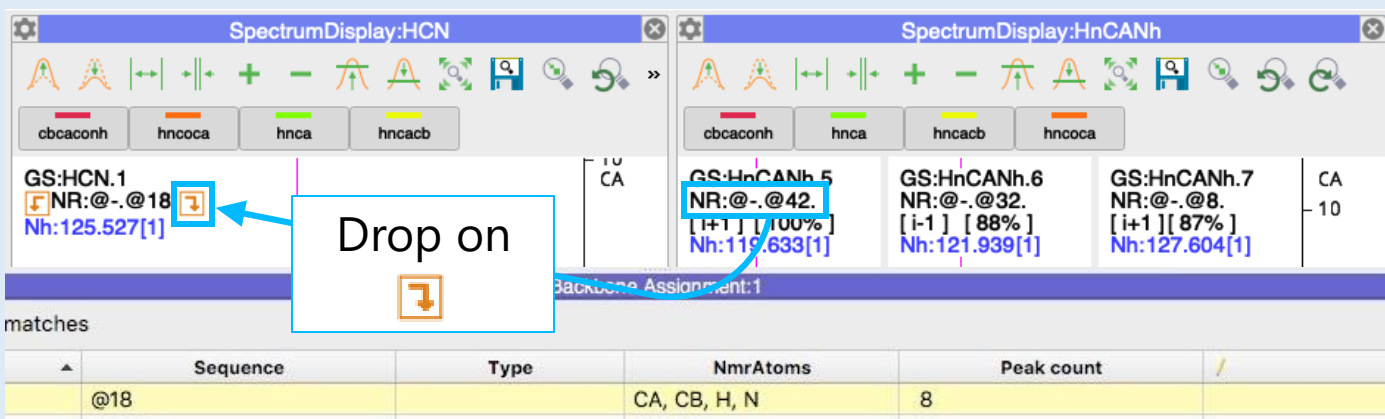
and **HNCLLTAEWMSAS** will be highlighted in the sequence because this is the best match in the sequence based on the chemical shifts.

- Select e.g. the label **#4.@56** in the Sequence Graph and drag it onto the left-hand residue (**N**) in the highlighted sequence
- Click **Yes** in the confirmation popup

You may notice a new NmrChain **A** has been automatically created and is linked to the Chain **A** for the above NmrResidues-Residues.

Now the relevant hsqc peaks for the selected stretch are fully assigned. You can inspect this in the **2D_HN** Spectrum Display.



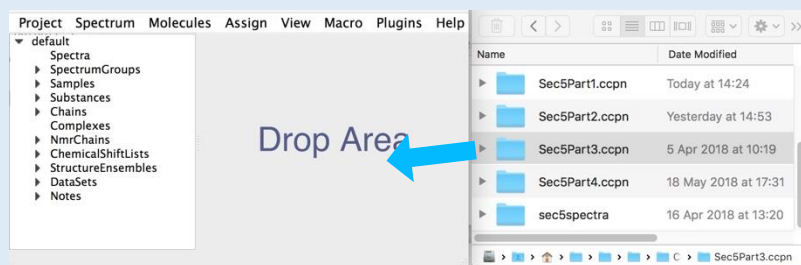


5G Assign in the i+1 direction

The Sequential backbone assignment in the i+1 direction is fully analogous to the 'i-1' direction except you will need to drop the strip label on the icon.

- Go to the Backbone Assignment module table
- Select chain @- (to display all nmrResidues again)
- Double-click on @18
- Go to the Match Spectrum Display **3D_HCN**
- Select the best match @42, drag & drop it onto the **3D_HCN_1** icon
- Continue in this direction to have a stretch consisting of: @18 @42 @34 @22 @99
- Select the @18 from the Sequence Graph, drag and drop onto the **L** of stretch **LTICG** highlighted in orange in the sequence below it.
- Click **Yes** in the confirmation popup.

This completes the steps required for a Sequential Backbone Assignment. The following parts of this tutorial are **How To's** and cover the usage of other Backbone Assignment tools, in particular how to inspect the assignment and edit it.

Open **Sec5Part3.ccpn**Open the Assignment Inspector with shortcut **AI**

Assignment Inspector

ChemicalShiftList CL:default

Value (ppm)	Value Error (ppm)	NmrAtom	ResidueType	AtomName	Peak Count	Total Peak Count
27.927	None	NA:A.19.THP.CB	THP	CB	1	1
29.667	None	NA:A.64.GLN.CB	GLN	CB	1	
59.188	3.346	NA:A.25-1..CA	None	CA	5	
66.475	1.780	NA:A.21-1..CA	None	CA	5	

Single click

Filter by NmrAtom(s): A.25-1..CA

#	Assign F1	Assign F2	Assign F3	Pos F1	Pos F2	Pos F3	LW F1
76	A.25.ARG.H	A.25-1..CA	A.25.ARG.N	8.597	52.496	124.006	20.823
75	A.25.ARG.H	A.25-1..CA	A.25.ARG.N	8.598	60.852	123.986	22.097
42	A.25.ARG.H	A.25-1..CA	A.25.ARG.N	8.597	60.860	123.999	21.756
22	A.25.ARG.H	A.25-1..CA	A.25.ARG.N	8.597	60.864	123.999	20.901
37	A.25.ARG.H	A.25-1..CA	A.25.ARG.N	8.596	60.868	123.993	21.719

Double click

Sec5Part3.ccpn is a project that contains a rather obvious assignment error; the NmrResidue 37-1 has a very high standard deviation (as calculated from all peaks). We will correct this mistake using Assignment Inspector and Peak Assigner modules.

6A Correcting assignment mistakes

- Open the project **Sec5Part3.ccpn**.
- Open the Assignment Inspector module:
 - Main Menu → Assign → Assignment Inspector** or shortcut **AI**
- Make sure the ChemicalShiftList **CL:default** is selected.
- **Click** on **Value Error** to sort the top table by the Chemical Shift error. Chemical Shifts with unusually high errors should be investigated.
- Select the row with NmrAtom **NA:A.25-1..CA**.
This will populate the peak table below with all peaks belonging to that NmrResidue (not NmrAtom!).
- Filter the peak table by clicking on **NA:A.25-1..CA** on the left hand side.
- **Click** on **Pos F2** to sort the peak table by the carbon chemical shift and find the outlier. It looks as though this peak must be incorrectly assigned.
- **Double-click** on row #76 in the peak table.
The DpectrumDisplay modules will automatically navigate to this peak position.

Current Peak: hncacb.1.76

H: 8.597 C: 52.496 N: 124.006

NmrAtom	Delta
A.25.ARG.H	0.001

NmrAtom	Delta
A.25-1..CA	6.692

NmrAtom	Delta
A.74.VAL.H	0.003

Alternatives

NmrAtom	Delta
A.26-1..CA	0.013
A.25.ARG.CA	0.039
A.50.ALA.CA	0.087
A.51-1..CA	0.089
A.65.ALA.CA	0.368
A.66-1..CA	0.379
A.67.ASN.CA	0.382
A.68-1..CA	0.390

Edit New

Current Peak: hncacb.1.76

H: 8.597 C: 52.496 N: 124.006

NmrAtom	Delta
A.25.ARG.H	0.001

NmrAtom	Delta
A.25.ARG.CA	0.039

NmrAtom	Delta
A.74.VAL.H	0.003

Alternatives

NmrAtom	Delta
A.26-1..CA	0.013
A.25.ARG.CA	0.039
A.50.ALA.CA	0.087
A.51-1..CA	0.089
A.65.ALA.CA	0.368
A.66-1..CA	0.379
A.67.ASN.CA	0.382
A.68-1..CA	0.390

Edit New

The **Peak Assigner** is very versatile and it has a number of options to guide its behaviour. Use the gear box icon to show/hide these. Crucial parameters are the assignment tolerances along each dimension for the different spectra. These are set to (sensible) default values but can also be modified by using the **Dimensions** tab of the **Spectrum Properties** popups of the respective spectra (open by double-clicking on a spectrum in the sidebar).

6B Edit the Assignment

While the peak which should be reassigned is selected in the table:

- Bring up the **Peak Assigner** module either by going to **Main Menu → View → Peak Assigner**, with shortcut **AP** or by **right-clicking** on your peak and selecting **Edit Peak Assignments**.

The upper panels show the peak assignments, the lower ones show NmrAtoms with Chemical Shifts close the peak position.

- Double-click** on **A.25-1..CA** in the top middle panel.

This will remove the incorrect assignment from the selected peak. The correct assignment is likely to be **A.25.ARG.CA**.

- Double-click** on **A.25.ARG.CA** in the lower middle panel to assign this NmrAtom to the selected peak.

Open Sec5Part4.ccpn

The screenshot shows the 'Sequence Graph' window in CCPN. The 'NmrChain' is set to 'NC:A'. The graph displays residues A.17.ILE, A.19.TRP, and A.20.THR. A right-click context menu is open over the 'H' atom of A.19.TRP. The menu includes options like 'deassign nmrAtoms from Peaks' and a list of assignments. The assignment 'A.69.LYS.N' is highlighted in the list. A blue arrow points from the text 'Right-click on H' to the 'H' atom, and another blue arrow points from the text 'Click to remove assignment' to the 'A.69.LYS.N' entry in the menu.

Sec5Part4.ccpn is a project that contains an assignment error; to identify assignment errors you can use the sequence graph. In this project, scrolling through the sequence graph of NmrChain A, you will notice multiple lines connecting the H atom of A.19.TRP to atoms much further down the sequence; in fact they go to the N of A.69.LYS. These are clearly mistakes in the assignment and should be corrected.

6c Correcting an assignment mistake

- Open the project **Sec5Part4.ccpn**.
- Open the **Sequence Graph** with shortcut **SG**.
- Select NmrChain **NC:A**.
- Scroll to the right and search for **A.19.TRP**. You will see immediately, that it has connections to an NmrResidue far away.
If you follow these you will see that the connections are to **A.69.LYS**.
- Select the **H**, **right-click** on it and look through each of the peaks assigned to this NmrAtom.
You will see all the assignments in the submenus. Find the incorrect assignments to **A.69.LYS.N**.
- **Click** on these incorrect assignments to remove them.

The wrong connections have now been removed, and the peaks have been de-assigned from the incorrect NmrAtoms. You can further edit the peak assignments to add the correct assignments using other tools like the Peak Assigner module (**AP**).

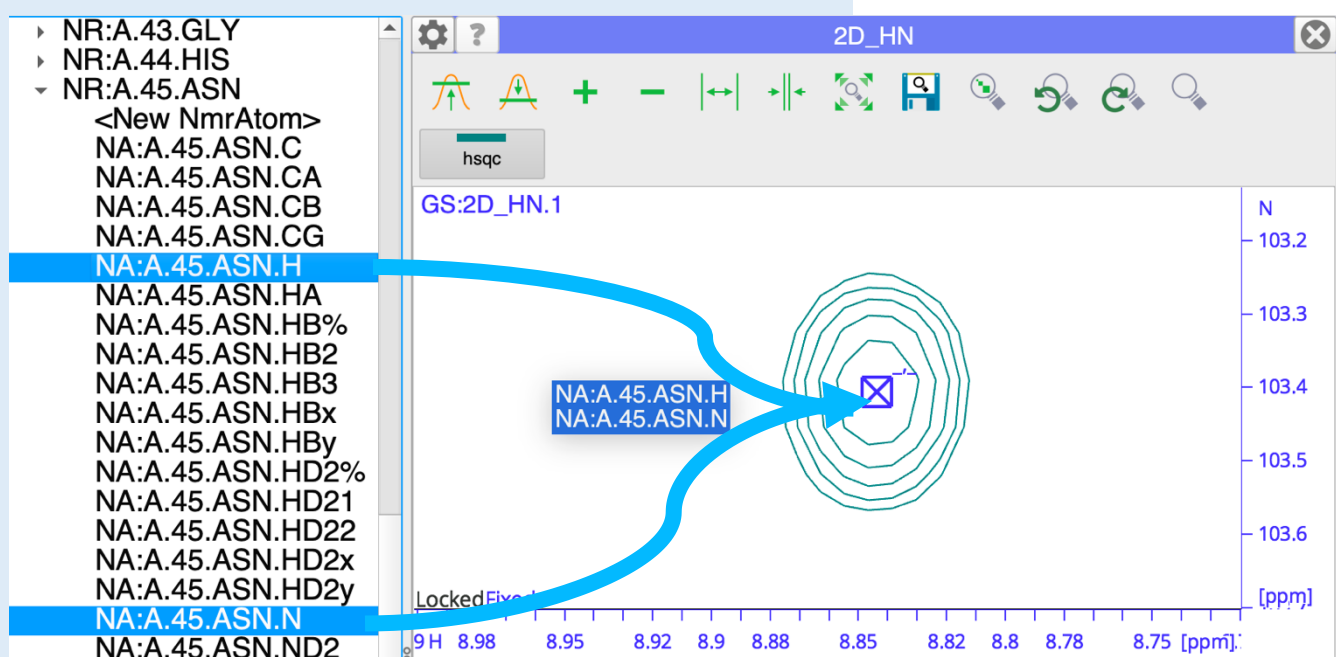
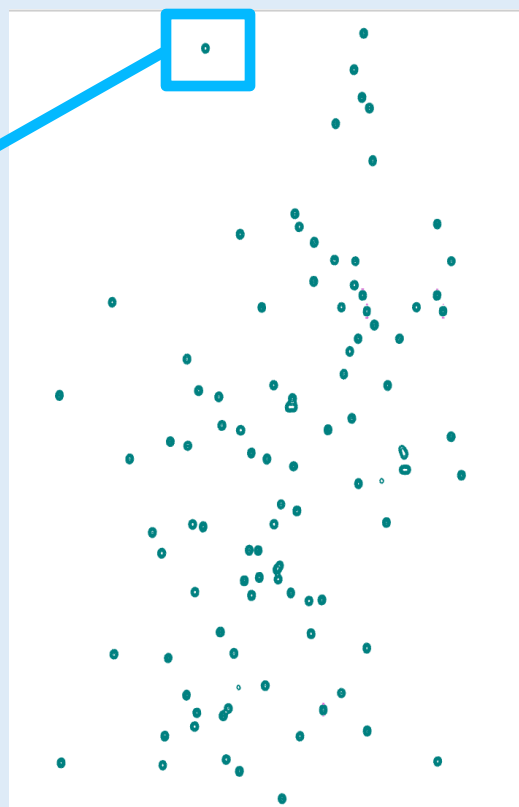
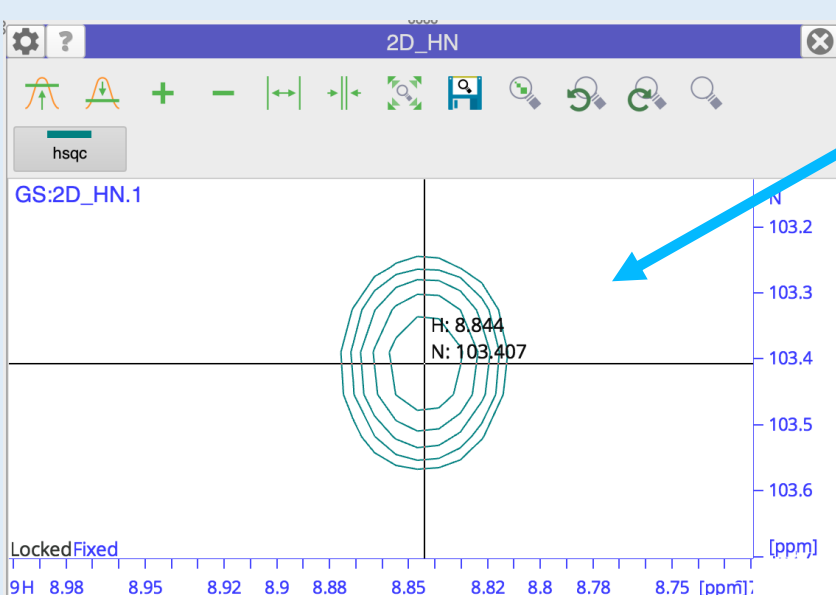
Have a look for other incorrect assignments and fix these, too, if you wish.

The image shows the AnalysisAssign software interface. The main window has a sidebar with a tree view under 'default' containing 'Spectra', 'SpectrumGroups', 'ChemicalShiftLists', 'NmrChains', 'Samples', 'Substances', 'Chains', 'Complexes', 'StructureEnsembles', 'StructureData', 'DataTables', 'Collections', and 'Notes'. A 'Drop Area' label with a blue arrow points to the main workspace. Another blue arrow points from a file explorer window to the same workspace. The file explorer shows two windows: 'sec5spectra' containing files 'cbcaconh.spc', 'cbcaconh.spc.par', 'hnca.spc', 'hsqc.spc', and 'hsqc.spc.par'; and 'CcpnBBAssignTutorial' containing '1hk6.fasta', 'README.txt', and 'Sec5Part1.ccpn'. A third screenshot shows a zoomed-in view of the 'NmrChains' section in the sidebar, with '<New NmrChain>' highlighted in blue. A blue arrow points from a 'Double click' label to this item. A fourth screenshot shows the 'Create NmrChain' dialog box with 'Create New' selected, 'Chain' chosen under 'Clone from', 'MC:A' in the dropdown, and 'A' in the 'Name' field. The 'Create' button is highlighted in blue.

Assuming you have pre-knowledge of a peak assignments, this HOW-TO will show you how to assign residues and atoms to any dimensionality peaks in few simple steps.

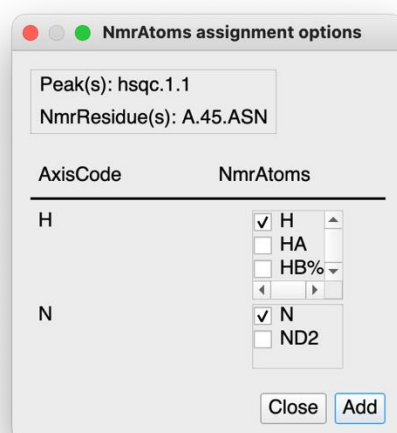
7_A Create a Chain and an NmrChain

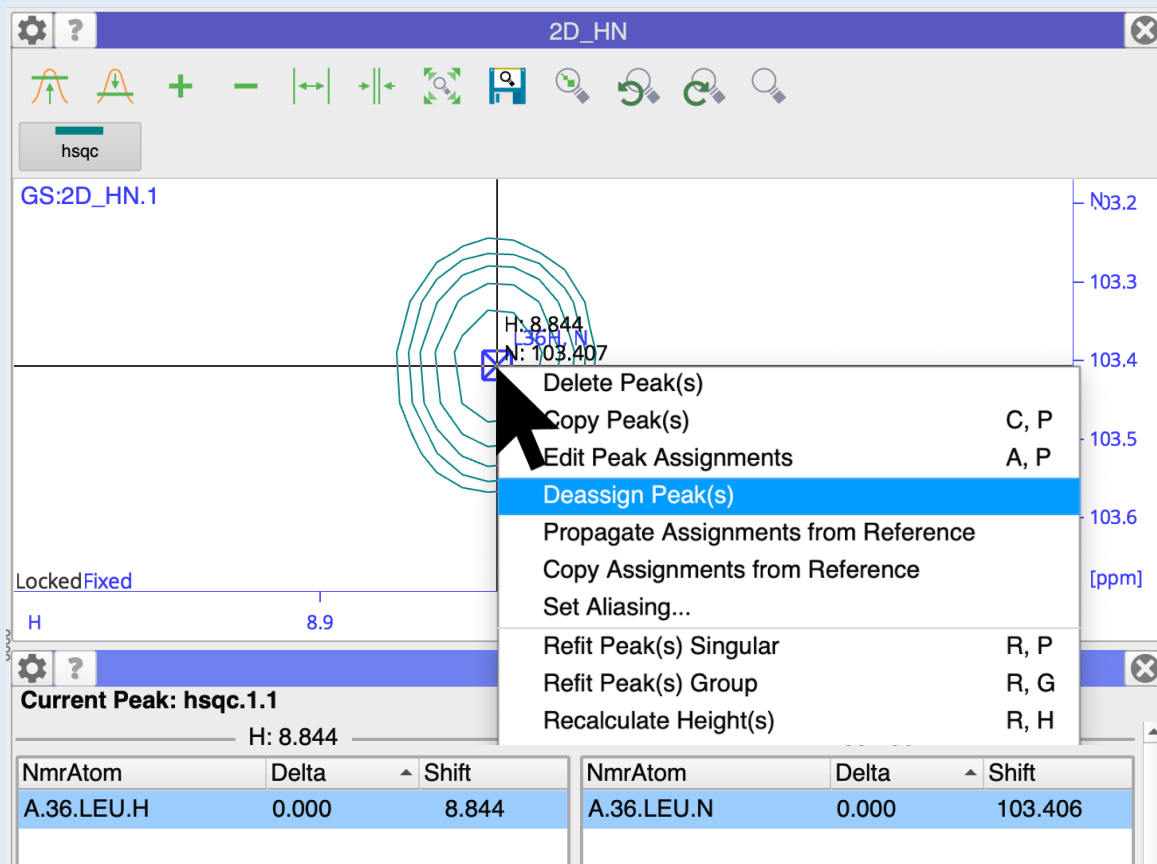
- Open a new project.
- **Drag and drop** the **hsqc.spc** spectrum from the **sec5spectra** folder into the Drop Area.
- **Drag and drop** the Fasta file **1hk6.fasta** into the Drop Area; this will automatically create a new Chain.
- In the sidebar, expand the **NmrChains** section and **double-click** on **<New NmrChain>**
- Select **Clone from: Chain**
- Select **Chain A** from the pulldown
- Name: A
- Click **Create**
- Open the hsqc in a SpectrumDisplay (**drag** into the Drop Area).

7_B**Drop the NmrAtoms or NmrResidue onto a selected peak**

- Zoom at positions: 8.84 / 103.40
- Pick the peak with **Ctrl (Cmd on Mac) + Shift (⇧) + right-click**
- In the sidebar expand the NmrChain **NC:A**
- Search for NmrResidue 45 (**NR:A.45.GLY**)
- Expand it and drag the H and N NmrAtoms onto the selected peak

You can also drag the NmrResidue **NR:A.45.GLY** and then select which atoms to assign to the peak:





You can assign any NmrAtoms to any dimensionality peaks as long as the NmrAtom's IsotopeCode matches that of one of the dimensions in the spectrum. E.g., you may assign **45.GLY.HA2** to any peak in an HN spectrum but you cannot assign the **45.GLY.O** to any of them.

7C Quick assignment edit

If you have made a mistake, then:

- You can deassign the peak by right-clicking on it and going to **Deassign Peak(s)**.
- Then reassign the peak or use the Peak Assigner to reassign it correctly.

Note: Dragging NmrAtoms or NmrResidues onto a Peak which has already been assigned will add additional assignments to the peak, not replace them.

Contact Us

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<https://forum.ccpn.ac.uk/>

Cite Us

Skinner, S. P. *et al.* CcpNmr AnalysisAssign: a flexible platform for integrated NMR analysis. J. Biomol. NMR 66, (2016)

Tutorial Version History:

beta1 (SS): First version

beta2 (GWV): Minor changes

beta3 (LGM): Re-designed, re-written, added several steps

V3 (VAH): Minor changes